

Question	Marking Guidance	Mark	Comments
01.1	1. Answer of 12/13 = 2 marks ;; 2. $0.36(48)/0.365/0.37 = 1 \text{ mark}$ OR $36(.48)/36.5/37\% = 1 \text{ mark}$ OR $q^2 = 0.06/0.059/0.0588 = 1 \text{ mark}$ OR or $q = 0.2/0.24/0.243 = 1 \text{ mark};$	2	For 1 mark accept $q^2 = 6\%/5.9\%/5.88\%$

Question	Marking Guidance	Mark	Comments
01.2	0.71	1	

Question	Marking Guidance	Mark	Comments
01.3	Second box ticked/answer key: B: The mutation that caused black fur happened in a common ancestor of <i>S. carolinensis</i> and other closely related species.	1	

Question	Marking Guidance	Mark	Comments
01.4	1. $2.55\% = 2 \text{ marks};;$ 2. $2.61\% = 1 \text{ mark}$ (question misread ie $8/306 \times 100$) OR Evidence of dividing by 314 or 942 = 1 mark OR Answers not given to three significant figures = 1 mark ;	2	

Question	Marking Guidance	Mark	Comments
01.5	1. Mutation/lack of glutamic acid leads to (permanent) activation of the receptor/protein; 2. (Because) the receptor/protein does not require the binding/leaving of αMSH (to become activated); 3. ASIP (might) not (be) able to bind to the receptor/protein; 4. (Only) the dark <u>pigment</u> is produced	3 max	2. Answer must convey the idea that binding/leaving is not required
